Selected Solutions from Tutorials

Adapted from **Shigley, Problem 17-2** (8th and 9th Eds., 17-1).

A 6-in-wide polyamide F-1 flat belt is used to connect a 2-in-diameter pulley to drive a larger pulley with an angular velocity ratio of 0.5. The centre-to-centre distance is 9 ft. The angular speed of the small pulley is 1750 rev/min as it delivers 2 hp. The application is such that a service factor K_s of 1.25 is appropriate.

Part 1: To be demonstrated by the TA.

a. Find F_c, F_i, F_{1a}, and F_2

d = 2 in (drive pulley diameter, given)
D = 2 in / 0.5 = 4 in (driven pulley diameter based on 0.5 velocity ratio)
b = 6in (belt width, given)

From Table 17-2 (p. 869) for a polyamide F-1 flat belt

t = 0.05 in (thickness)
$\gamma = 0.035$ lbf/in^3 (specific weight)
f = 0.5 (coefficient of friction)
F_a = 35 lb/in (allowable tension per width at 600 ft/min)

F_c = hoop tension due to centrifugal force

$$F_c = \frac{w}{g}\left(\frac{V}{60}\right)^2 = \frac{w}{32.17}\left(\frac{V}{60}\right)^2$$ (eq. 17-8e, p. 885)

$$V = \pi dn/12$$
$$= \pi(2\text{ in})(1750\text{ rev/min})/12$$ (belt speed in ft/min, p. 885)
$$= 916.3\text{ ft/min}$$

$$w = 12\gamma bt$$
$$= 12\left(0.035 \text{ lb/in}^3\right)(6 \text{ in})(0.05 \text{ in}) \qquad \text{(belt weight per foot, p. 885)}$$
$$= 0.126 \text{ lb/ft}$$

$$F_c = \frac{0.126 \text{ lb/ft}}{32.17}\left(\frac{916.3 \text{ ft/s}}{60}\right)^2$$
$$= 0.913 \text{ lb}$$

F_c = 0.913 lb

F_{1a} = allowable largest tension

$F_{1a} = F_a b C_p C_v$ (eq. 17-12, p. 888)

F_a = 35 lb/in (allowable tension per width, Table 17-2, p. 889)
b = 6in (belt width, given)

C_p = 0.70 (Table 17-4, p. 890, with 2in small pulley diameter)

C_v = 1 (velocity correction factor, =1 for polyamide and urethane belts, see note on p. 887)

F_{1a} = (35 lb/in)(6 in)(0.70)(1.0)
= 147 lb

F_{1a} = 147 lb

F_2 = loose side tension

$F_2 = F_{1a} - \Delta F$
$= F_{1a} - 2T/d$

Need to find T

$T = H_d/\omega$ (torque transmitted = power supplied / angular velocity)

$H_d = H_{nom} K_s n_d$ (see note on p. 888
H_d = power delivered
H_{nom} = nominal power delivered = 2 hp in this case, given
K_s = service factor (for deviations from normal load)
= 1.25 in this case, given
n_d = design safety factor = 1 in this case, implied)

$T = H_d/\omega$ is in SI units, in units of hp, lb·in, and rev/min the expression is

$$T = \frac{63{,}025H_d}{n}$$
$$= \frac{63{,}025H_{nom}K_s n_d}{n}$$
$$= \frac{63{,}025(2\text{ hp})(1.25)(1)}{1750\text{ rev/min}}$$
$$= 90.0\text{ lb}\cdot\text{in}$$

F_2 = F_{1a} - 2T/d
= 147 lb - 2(90.0 lb·in)/(2 in)
= 57 lb

F_2 = 57 lb

F_i = initial tension

F_i = 0.5 (F_1 + F_2) - F_c (eq. 17-8j, p. 886)
= 0.5 (147 lb + 57 lb) - 0.916 lb
= 101.1 lb

F_i = 101.1 lb

Part 2: To be done by students.

b. Find the allowable power, H_a

H_a = transmitted power with maximum allowable tension used
H_a = (F_{1a} - F_2)·V / 33,000 (from eq. (j), p. 887, note: the 33,000 is so units work in hp, lb, and ft/min)

= (147 lb - 57 lb)·(916.3 ft/min)/33,000
= 2.5 hp

H_a = 2.5 hp

c. Find the factor of safety, n_{fs}

Note: you may need to explain in words how to compute the factor of safety

n_{fs} = (allowable power) / [(nominal power)·(service factor)]
= H_a / [H_{nom}·K_s]
= (2.5 hp) / [(2 hp)(1.25)] (transmitted power found above,
= 1

n_{fs} = 1

d. Find the belt length, L

$$L = \sqrt{4C^2 - (D-d)^2} + \frac{1}{2}(D\theta_D + d\theta_d)$$ (eq. 17-2, p. 883)

C = centre distance = 9 ft (given)
d = small pulley diameter = 2in (given)
D = large pulley diameter = 4 in (found above, d/0.05)

$$\theta_d = \pi - 2\sin^{-1}\frac{D-d}{2C}$$
$$= \pi - 2\sin^{-1}\left(\frac{4\text{ in} - 2\text{ in}}{2 \cdot 9\text{ ft} \cdot 12\text{ in/ft}}\right)$$ (eq. 17-1, p. 883, must use radians)
$$= 3.132\text{ rad}$$

$$L = \sqrt{4(9\text{ ft} \cdot 12\text{ in/ft})^2 - (4\text{ in} - 2\text{ in})^2} + \frac{1}{2}\left[(4\text{ in})(3.151) + (2\text{ in})(3.132)\right]$$
$$= 225.4\text{ in}$$

L = 225.4 in

e. Find the dip

$$dip = \frac{C^2 w}{96 F_i}$$
$$= \frac{(9\text{ ft} \cdot 12\text{ in/ft})^2 (0.126\text{ lb/ft})}{96(101.1\text{ lb})}$$ (e.q. 17-13, p. 892)
$$= 0.151\text{ in}$$

(Note: the mismatched units are accounted for in the coefficient 96; see p. 892 for details.)

Dip = 0.151 in

A polyamide A-3 flat belt 6 in wide is used to transmit 15 hp under light shock conditions where $K_s = 1.25$, and a factor of safety equal to or greater than 1.1 is appropriate. The pulley rotational axes are parallel and in the horizontal plane. The shafts are 8 ft apart. The 6-in driving pulley rotates at 1750 rev/min in such a way that the loose side is on top. The driven pulley is 18 in in diameter. See Fig. 17–10. The factor of safety is for unquantifiable exigencies.

(*a*) Estimate the centrifugal tension F_c and the torque T.

(*b*) Estimate the allowable F_1, F_2, F_i and allowable power H_a.

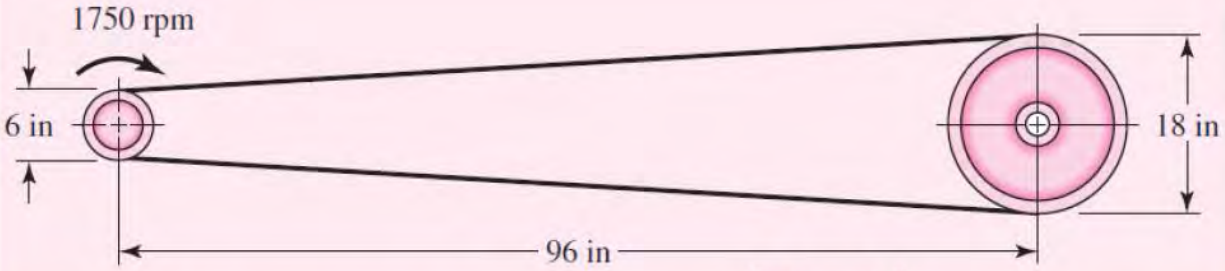
(*c*) Estimate the factor of safety. Is it satisfactory?

Belt 6 in × 0.130 in

15 hp

$\gamma = 0.042 \ \frac{\text{lbf}}{\text{in}^3}$

$d = 6$ in, $D = 18$ in



(*a*) Eq. (17–1): $$\phi = \theta_d = \pi - 2\sin^{-1}\left[\frac{18-6}{2(8)12}\right] = 3.0165 \text{ rad}$$

Table 17–2: $\gamma = 0.042$ lbf/in^3 $\quad f = 0.8$ $\quad F_a = 100$ lbf/in

$$\exp(f\phi) = \exp[0.8(3.0165)] = 11.17$$

$$V = \pi(6)1750/12 = 2749 \text{ ft/min}$$

$$w = 12\gamma bt = 12(0.042)6(0.130) = 0.393 \text{ lbf/ft}$$

Eq. (*e*): $$F_c = \frac{w}{g}\left(\frac{V}{60}\right)^2 = \frac{0.393}{32.17}\left(\frac{2749}{60}\right)^2 = 25.6 \text{ lbf}$$

$$T = \frac{63\,025 H_{\text{nom}} K_s n_d}{n} = \frac{63\,025(15)1.25(1.1)}{1750}$$

$$= 742.8 \text{ lbf} \cdot \text{in}$$

(*b*) The necessary $(F_1)_a - F_2$ to transmit the torque T, from Eq. (*h*), is

$$(F_1)_a - F_2 = \frac{2T}{d} = \frac{2(742.8)}{6} = 247.6 \text{ lbf}$$

For polyamide belts $C_v = 1$, and from Table 17–4 $C_p = 0.70$. From Eq. (17–12) the allowable largest belt tension $(F_1)_a$ is

$$(F_1)_a = bF_aC_pC_v = 6(100)0.70(1) = 420 \text{ lbf}$$

then

$$F_2 = (F_1)_a - [(F_1)_a - F_2] = 420 - 247.6 = 172.4 \text{ lbf}$$

and from Eq. (*i*)

$$F_i = \frac{(F_1)_a + F_2}{2} - F_c = \frac{420 + 172.4}{2} - 25.6 = 270.6 \text{ lbf}$$

The combination $(F_1)_a$, F_2, and F_i will transmit the design power of $H_a = H_{\text{nom}}K_sn_d = 15(1.25)(1.1) = 20.6$ hp and protect the belt. We check the friction development by solving Eq. (17–7) for f':

$$f' = \frac{1}{\phi}\ln\frac{(F_1)_a - F_c}{F_2 - F_c} = \frac{1}{3.0165}\ln\frac{420 - 25.6}{172.4 - 25.6} = 0.328$$

As determined earlier, $f = 0.8$. Since $f' < f$, there is no danger of slipping.

(*c*) From step 9 on p. 880,

$$n_{fs} = \frac{H_a}{H_{\text{nom}}K_s} = \frac{20.6}{15(1.25)} = 1.1 \qquad \text{(as expected)}$$

The belt is satisfactory and the maximum allowable belt tension exists. If the initial tension is maintained, the capacity is the design power of 20.6 hp.

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1. Adapted from Shigley, Problem 17-26.

A four-strand no. 40 roller chain transmits power from a 21-tooth driving sprocket to an 84-tooth driven sprocket. The angular speed of the driving sprocket is 2000 rpm.

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To be done by the students.

a. If the centre-to-centre distance needs to be about 20 inches, what is the chain length?

Given: No. 40-4 chain, N_1 = 21 teeth for n = 2000 rev/min, N_2 = 84 teeth

Chain pitch is p = 4/8 = 0.500 inches and C ≈ 20 inches.

Eq. (17-34):

$$\frac{L}{p} = \frac{2C}{p} + \frac{N_1 + N_2}{2} + \frac{(N_1 - N_2)^2}{4\pi^2 C / p}$$

$$= \frac{2(20)}{0.5} + \frac{21 + 84}{2} + \frac{(84 - 21)^2}{4\pi^2 (20 / 0.5)} = 135 \text{ pitches} \quad \text{(or links)}$$

L = 135(0.500) = 67.5 in *Ans.*

L = 67.5 in

b. Determine the tabulated horsepower adjusting for K_1 and K_2.

Table 17-20: No.40 chain, at 2000 rpm, H_{tab} = 7.72 hp

Table 17-22: Note that this is now the "post-extreme" value, that is, failure by fatigue of the roller (occurs after the highest value for H_{Tab}),

$$K_1 = \left(\frac{N_1}{17}\right)^{1.5} = \left(\frac{21}{17}\right)^{1.5} = 1.37$$

Table 17-23: Four strands, K_2 = 3.3

Eq. (17-37): $H_a = K_1K_2H_{tab} = 1.37(3.3)7.72 = 34.9$ hp *Ans.*

c. Estimate the allowable horsepower from Equations (17-32) and (17-33) (which are the sources for Table 17-20) and compare with the value found in part (b). Comment on the result.

Using Eq. (17-32), the link plate limiting equation:

$$H_1 = 0.004N_1^{1.08}n_1^{0.9}p^{(3-0.07p)} = 0.004(21)^{1.08}(2000)^{0.9}(0.5)^{[3-0.07(.5)]} = 12.84 \text{ hp}$$

To compare with the tabulated value from part (b), adjust with K_2 for the number of strands. It is not necessary to adjust with K_1, since the number of teeth is included in Eq. (17-32).

$$H_a = K_2H_1 = 3.3(12.84) = 42.4 \text{ hp}$$

This is larger than the table value from part (b), probably indicating that H_2 (the roller failure) is the limiting power in the table.

Eq. (17-33), the roller/pin limiting equation:

$$H_2 = \frac{1000K_rN_1^{1.5}p^{0.8}}{n_1^{1.5}} = \frac{1000(17)(21)^{1.5}0.5^{0.8}}{(2000)^{1.5}} = 10.5 \text{ hp}$$

Adjusting for the number of strands,

$H_a = K_2H_2 = 3.3(10.5) = 34.7$ hp *Ans.*

This compares pretty well to the adjusted table value from part (b) of 34.9 hp. Hence we can predict the failure will occur in the roller/pin and not the link plate. Note that the asterisks value in the Table states that these H_{Tab} are found by a linear interpolation of the data found in the ANSI tables and not the equations.

To be demonstrated by the TA.

d. Determine the allowable horsepower for a 20,000 hour life for this chain drive system.

Eq. (17-39) gives the roller/pin limiting state for deviations from the normal test conditions (15,000 hours, 17 teeth, 100 pitch length) as:

$$H_2 = 1000\left[K_r\left(\frac{N_1}{n_1}\right)^{1.5} p^{0.8}\left(\frac{L_p}{100}\right)^{0.4}\left(\frac{15000}{h}\right)^{0.4}\right]$$

$$= 1000\left[17\left(\frac{21}{2000}\right)^{1.5} 0.5^{0.8}\left(\frac{135}{100}\right)^{0.4}\left(\frac{15000}{20000}\right)^{0.4}\right] = 10.56 \text{ hp}$$

Note: K_r in equation 17-39 is a constant. It turns out that K_r = 17 which is the number of teeth for the standard sprocket used in the endurance tests. The value of K_r is NOT indicated in the Shigley text. Students may wish to note it in their text.

Adjusting for the 4 strands,

$$H_a = K_2 H_2 = 3.3(10.56) = 34.8 \text{ hp} \qquad \textit{Ans.}$$

This is only slightly different from the 34.9 and 34.7 hp from parts (b) and (c). Note that accounting for the longer chain length increased the power rating slightly, while accounting for the longer life decreased the power rating slightly.

e. Estimate the tension in the chain at the allowable horsepower.

From the standard relationship of power, torque, and speed, $H=T\omega$

$$T = \frac{H}{\omega} = \frac{34.8 \text{ hp}}{2000 \text{ rev/min}} \left(\frac{550 \text{ ft-lbf/s}}{\text{hp}} \right) (60 \text{ s/min}) \left(\frac{1}{2\pi \text{ rad/rev}} \right) = 91.4 \text{ lbf-ft}$$

Eq. (17-29): $$D = \frac{p}{\sin(180^\circ / N)} = \frac{0.5}{\sin(180^\circ / 21)} = 3.355 \text{ in}$$

$$F_1 = T / (D/2) = 91.4 \text{ lbf-ft} (12 \text{ in/ft}) / \left[(3.355/2) \text{ in} \right]$$
$$= 654 \text{ lbf} \qquad \textit{Ans.}$$

Chain Tension = 654 lbf.

1. Adapted from Shigley, Problem 17-19.

A 60 hp four cylinder internal combustion engine is used to drive a medium-shock brick-making machine under a schedule of 2 shifts per day. The drive consists of two 26 inch sheaves spaced about 12 feet apart, with a sheave speed of 400 rpm. Select a Gates Rubber V-belt D360 arrangement.

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a. Determine the required number of belts.

Given: H_{nom} = 60 hp, n = 400 rev/min, $d = D$ = 26 in on 12 ft centers.

From Table 17-15, a conservative service factor of K_S = 1.4 is selected.

Design task: specify V-belt and number of strands (belts). *Tentative decision*: Use D360 belts.

From Table 17-11: L_p = 360 + 3.3 = 363.3 in

FromEq.(17-16*b*):

$$C = 0.25\left\{\left[363.3 - \frac{\pi}{2}(26 + 26)\right] + \sqrt{\left[363.3 - \frac{\pi}{2}(26 + 26)\right]^2 - 2(26 - 26)^2}\right\}$$

$$= 140.8 \text{ in (nearly 144 in)}$$

$$\theta_d = \pi, \quad \theta_D = \pi, \quad \exp[0.5123\pi] = 5.0,$$

$$V = \frac{\pi dn}{12} = \frac{\pi(26)(400)}{12} = 2722.7 \text{ ft/min}$$

Table 17-13: For θ = 180°, K_1 = 1

Table 17-14: For D360, K_2 = 1.10

Table 17-12: H_{tab} = 16.94 hp by interpolation

Thus Eq. (17-17), $H_a = K_1K_2H_{tab} = 1(1.1)(16.94) = 18.63$ hp / belt

Eq. (17-19): $H_d = H_{nom}K_s\, n_d = 60(1.4)(1) = 84$ hp

Number of belts, N_b

$$N_b = \frac{H_d}{H_a} = \frac{84}{18.63} = 4.51$$

Round up to five belts.

A total of five D360 belts are required.

b. What is the factor of safety?

From Eq. (17-22) the transmitted force and torque per belt is:

$$\Delta F_a = \frac{63\,025H_d}{n(d/2)} = \frac{63\,025(18.63)}{400(26/2)} = 225.8 \text{ lbf/belt}$$

$$T_a = \frac{(\Delta F_a)d}{2} = \frac{225.8(26)}{2} = 2935 \text{ lbf} \cdot \text{in/belt}$$

The centripetal tension Eq. (17-21) is:

$$F_c = 3.498\left(\frac{V}{1000}\right)^2 = 3.498\left(\frac{2722.7}{1000}\right)^2 = 25.9 \text{ lbf/belt}$$

At fully developed friction, Eq. (17-9) gives

$$F_i = \frac{T}{d}\left[\frac{\exp(f\theta)+1}{\exp(f\theta)-1}\right] = \frac{2935}{26}\left(\frac{5+1}{5-1}\right) = 169.3 \text{ lbf/belt}$$

Eq. (17-10): $$F_1 = F_c + F_i\left[\frac{2\exp(f\theta)}{\exp(f\theta)+1}\right] = 25.9 + 169.3\left[\frac{2(5)}{5+1}\right] = 308.1 \text{ lbf/belt}$$

$$F_2 = F_1 - \Delta F_a = 308.1 - 225.8 = 82.3 \text{ lbf/belt}$$

$$n_{fs} = \frac{H_a N_b}{H_d} = \frac{18.63(5)}{84} = 1.109 \quad \textit{Ans.}$$

Safety Factor = 1.1. This safety factor is barely enough. One could increase the safety factor by adding more belts or moving to a higher rated belt (type E).

c. Estimate the life in passes, hours, and days.

From Table 17-16,

$$T_1 = T_2 = F_1 + \frac{K_b}{d} = 308.1 + \frac{5\,680}{26} = 526.6 \text{ lbf}$$

From Table 17-17, for D-belts, K=4208 and b=11.105

Using Eq. (17-27):

$$N_P = \left[\left(\frac{K}{T_1}\right)^{-b} + \left(\frac{K}{T_2}\right)^{-b}\right]^{-1} = 5.28\left(10^{-9}\right) \text{ passes (Note } 10^{9} \text{ not } 10^{-9})$$

Thus, $N_P > 10^9$ passes *Ans.*

Eq. (17-28): $t = \dfrac{N_P L_p}{720V} > \dfrac{10^9(363.3)}{720(2722.7)}$

Thus, t > 185 320 h *Ans*.

For two shifts per day we have 16 operating hours/day so

days > 185 320 h/16 h/day = 11582.5 days/365.25 = 31.7 years *Ans*.

The expected life is greater than 10^9 passes and 31 years. This would justify the above safety factor.

2. Re-do Problem 17-19 but use Gates V-Belt Design Manual

A 60 hp four cylinder internal combustion engine is used to drive a brick-making machine under a schedule of 2 shifts per day. The drive consists of two equal diameter sheaves spaced about 12 feet apart, with a sheave speed of 400 rpm.

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a) Determine required number of belts.

Here you will need to refer to the Gates Design manual to solve the problem. The manual is found on Canvas or downloaded from the Gates website. You can use the procedure outlined on Page B2

Given: H_{nom} = 60 hp, n = 400 rev/min, $d = D$ on 12 ft centers.

From Table B1, a service factor of K_S = 1.5 is selected. We have a multiple cylinder engine operating > 10hrs/day for Brick Machinery.

Table No. B1 — Service Factors

DriveN Machine	DriveR					
The machines listed below are representative samples only. Select the group listed below whose load characteristics most closely approximate those of the machine being considered.	**AC Motors:** Normal Torque, Squirrel Cage, Synchronous, Split Phase. **DC Motors:** Shunt Wound. **Engines:** Multiple Cylinder Internal Combustion.*			**AC Motors:** High Torque, High Slip, Repulsion-Induction, Single Phase, Series Wound, Slip Ring. **DC Motors:** Series Wound, Compound Wound. **Engines:** Single Cylinder Internal Combustion.* **Line shafts** **Clutches**		
	Intermittent Service	Normal Service	Continuous Service	Intermittent Service	Normal Service	Continuous Service
	3-5 Hours Daily or Seasonal	8-10 Hours Daily	16-24 Hours Daily	3-5 Hours Daily or Seasonal	8-10 Hours Daily	16-24 Hours Daily
Dispensing, Display Equipment Instrumentation Measuring Equipment Medical Equipment Office, Projection Equipment	1.0	1.1	1.2	1.1	1.2	1.3
Agitators: Liquid Appliances, Sewing Machines, Sweepers Conveyors: Belt, Light Package Fans: Up to 10 HP Hand Tools (Power) Machine Tools: (Light) Drill Presses, Lathes, Saws Screens: Drum, Oven Woodworking Equipment: Band Saws, Drills, Lathes	1.1	1.2	1.3	1.2	1.3	1.4
Agitators: Semi-liquid Compressors: Centrifugal Centrifuges Conveyors: Belt, Coal, Ore, Sand Dough Mixers Fans: Over 10 HP Generators Laundry Equipment Line Shafts Machine Tools: (Heavy) Boring, Grinders, Milling, Shapers Paper Machinery (except Pulpers) Presses, Punches, Shears Printing Machinery Pumps: Centrifugal, Gear Screens: Revolving, Vibratory	1.1	1.2	1.4	1.2	1.3	1.5
Blowers: Positive Displacement, Mine Fans Brick Machinery Compressors: Piston Conveyors: Drag, Elevator, Pan, Screw Elevators: Bucket Exciters Extractors Mills: Hammer Paper Pulpers Pulverizers Pumps: Piston Rubber Calendars, Extruders, Mills Textile Machinery	1.2	1.3	1.5	1.4	1.5	1.6
Crushers (Gyratory-Jaw-Roll) Hoists Mills: Ball-Rod-Tube Sawmill Machinery	1.3	1.4	1.6	1.5	1.6	1.8

*Apply indicated Service Factor to continuous engine rating. Deduct 0.2 (with a minimum Service Factor of 1.0) when applying to maximum intermittent rating. The use of a Service Factor of 2.0 is recommended for equipment subject to choking. For Grain Milling and Elevator Equipment, see Mill Mutual Bulletin No. VB-601-62. For Oil Field Machinery, see API specification for Oil Field V-Belting, API Standard 1B.

Hence the design power is (60)(1.5) = 90HP

The next step will be to select a belt type using the selection charts (Page B3). For a design power of 90HP at 400rpm, two options are available. Series 8V or 8VX and series CP. Let's choose the Hi Power CP series as they are designed for rugged operations (See page A7). These are the A,B,C and D family of belts.

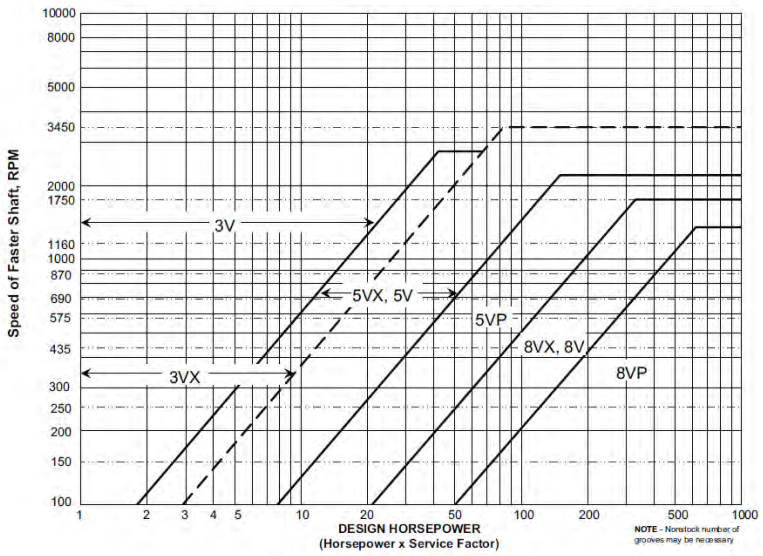


Figure No. B1

Cross Section Selection Chart

(For Super HC® V-Belts, Super HC® Molded Notch V-Belts, Super HC PowerBand Belts, and Predator Belts)

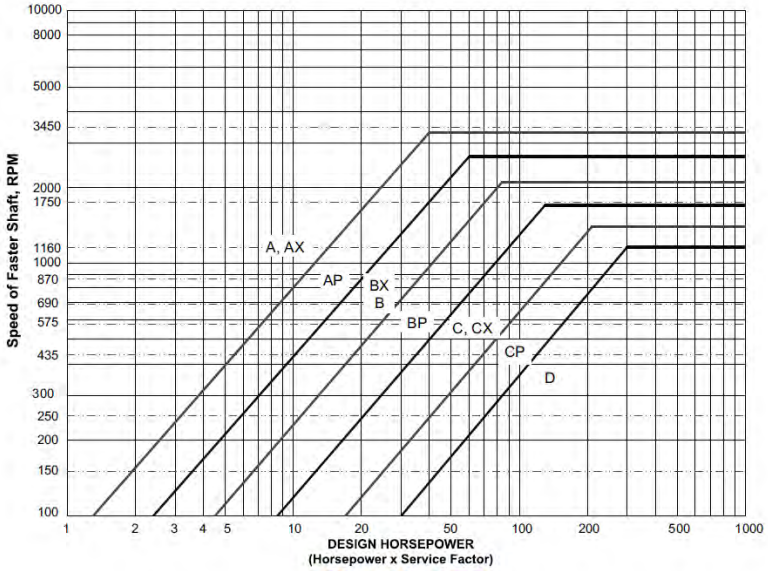


Figure No. B2

Cross Section Selection Chart

(For Hi-Power® II V-Belts, Hi-Power II PowerBand Belts, Tri-Power® Molded Notch V-Belts, and Predator Belts)

Predator® Single Belts

Specifically designed for aggressive applications, this extra heavy-duty belt construction provides extraordinary high impact strength, capacity, and wear resistance.

Markets/Applications

Predator belts are well suited as replacement belts for applications exposed to pulsating loads or heavy shock loads, such as mining, agriculture, wood processing, oil field equipment, heavy construction and sand/gravel operations.

Features/Advantages

- Aramid Tensile Cords combine limited stretch with extraordinary strength and durability that is pound-for-pound stronger than steel.
- Bareback Cover resists debris and allows the belt to slip un-der extreme shock load conditions, reducing heat buildup and prolonging belt life.
- Gates Curves provide proper cord support and full contact with the sheave-groove for equal loading, uniform wear, and increased belt life.
- Meets RMA oil and heat resistant standards.



The next step is identify the minimum Sheave Diameter for the motor. This is found using the Tables on page B4. Be sure to select the correct Table for the type of belt. Since our belt is from the Hi Power family (Figure B2) we use Table B3.For 60 HP and 575 rpm, value is 12 inches. Given our 400 rpm, our sheave should exceed 12".

Table No. B3

Minimum Recommended Sheave Datum Diameters for General Purpose Electric Motors
Hi-Power® II V-Belts, Hi-Power II PowerBand Belts or Tri-Power® Molded Notch V-Belts

** For U.S. Only

Motor Horse-power	Motor RPM (60 cycle and 50 cycle Electric Motors)						Motor Horse-power
	575 485*	690 575*	870 725*	1160 950*	1750 1425*	3450 2850*	
½	2.5	2.5	2.2	—	—	—	½
¾	3.0	2.5	2.4	2.2	—	—	¾
1	3.0	3.0	2.4	2.4	2.2	—	1
1½	3.0	3.0	2.4	2.4	2.4	2.2	1½
2	3.8	3.0	3.0	2.4	2.4	2.4	2
3	4.5	3.8	3.0	3.0	2.4	2.4	3
5	4.5	4.5	3.8	3.0	3.0	2.6	5
7½	5.2	4.5	4.4	3.8	3.0	3.0	7½
10	6.0	5.2	4.6	4.4	3.8	3.0	10
15	6.8	6.0	5.4	4.6	4.4	3.8	15
20	8.2	6.8	6.0	5.4	4.6	4.4	20
25	9.0	8.2	6.8	6.0	5.0	4.4	25
30	10.0	9.0	6.8	6.8	5.4	—	30
40	10.0	10.0	8.2	6.8	6.0	—	40
50	11.0	10.0	9.0	8.2	6.8	—	50
60	12.0	11.0	10.0	9.0	7.4	—	60
75	14.0	13.0	10.5	10.0	9.0	—	75

Next we want to determine the belt size for our application. We are given a center distance of 12 feet or 144 inches.
We go to the Belt Selection Tables to find the correct belt for application. Here you need to demonstrate the numerous options available and remind students to find the correct "family" of belts. We are looking for the Hi Power CP Belts. These class of belts start at page B67. The C Belts begin at page B184.

We are looking for a Speed Ratio of 1.0 (Column 5), Sheave size larger than 12 and a Center Distance of approx. 144 inches. Scroll to page B191 to find the required belt.

Table No. B23

Hi-Power® II V-Belt, Tri-Power® Molded Notch V-Belt, Predator V-Belt, Hi-Power® II PowerBand® Belt and Predator PowerBand Belt Drives



V-Belt No. and Center Distance																							Speed Ratio	Sheave Datum Diameters	
C 250	C CX CP 255	C 264	C 265	C CX 270	C 275	C 276	C 280	C 285	C 290	C 295	C 297	C CX 300	C 303	C 314	C 315	C 320	C CX CP 330	C CP 345	C CX CP 360	C 390	C 420	C 450		Small Sheave	Large Sheave
	120.1			127.6								142.6					157.6		172.6				1.00	* 5.00	* 5.00
	119.2			126.7								141.7					156.7		171.7				1.00	* 5.60	* 5.60
	118.5			126.0								141.0					156.0		171.0				1.00	* 6.00	* 6.00
	117.0			124.5								139.5					154.5		169.5				1.00	* 7.00	* 7.00
	116.2			123.7								138.7					153.7		168.7				1.00	* 7.50	* 7.50
	115.4			122.9								137.9					152.9		167.9				1.00	* 8.00	* 8.00
	114.6			122.1								137.1					152.1		167.1				1.00	* 8.50	* 8.50
111.3	113.8	118.3	118.8	121.3	123.8	124.3	126.3	128.8	131.3	133.8	134.8	136.3	137.8	143.3	143.8	146.3	151.3	158.8	166.3	181.3	196.3	211.3	1.00	9.00	9.00
110.5	113.0	117.5	118.0	120.5	123.0	123.5	125.5	128.0	130.5	133.0	134.0	135.5	137.0	142.5	143.0	145.5	150.5	158.0	165.5	180.5	195.5	210.5	1.00	9.50	9.50
109.7	112.2	116.7	117.2	119.7	122.2	122.7	124.7	127.2	129.7	132.2	133.2	134.7	136.2	141.7	142.2	144.7	149.7	157.2	164.7	179.7	194.7	209.7	1.00	10.00	10.00
109.0	111.5	116.0	116.5	119.0	121.5	122.0	124.0	126.5	129.0	131.5	132.5	134.0	135.5	141.0	141.5	144.0	149.0	156.5	164.0	179.0	194.0	209.0	1.00	10.50	10.50
108.2	110.7	115.2	115.7	118.2	120.7	121.2	123.2	125.7	128.2	130.7	131.7	133.2	134.7	140.2	140.7	143.2	148.2	155.7	163.2	178.2	193.2	208.2	1.00	11.00	11.00
106.6	109.1	113.6	114.1	116.6	119.1	119.6	121.6	124.1	126.6	129.1	130.1	131.6	133.1	138.6	139.1	141.6	146.6	154.1	161.6	176.6	191.6	206.6	1.00	12.00	12.00
105.0	107.5	112.0	112.5	115.0	117.5	118.0	120.0	122.5	125.0	127.5	128.5	130.0	131.5	137.0	137.5	140.0	145.0	152.5	160.0	175.0	190.0	205.0	1.00	13.00	13.00
103.5	106.0	110.5	111.0	113.5	116.0	116.5	118.5	121.0	123.5	126.0	127.0	128.5	130.0	135.5	136.0	138.5	143.5	151.0	158.5	173.5	188.5	203.5	1.00	14.00	14.00
100.3	102.8	107.3	107.8	110.3	112.8	113.3	115.3	117.8	120.3	122.8	123.8	125.3	126.8	132.3	132.8	135.3	140.3	147.8	155.3	170.3	185.3	200.3	1.00	16.00	16.00
110.9	113.4	117.9	118.4	120.9	123.4	123.9	125.9	128.4	130.9	133.4	134.4	135.9	137.4	142.9	143.4	145.9	150.9	158.4	165.9	180.9	195.9	210.9	1.05	9.00	9.50
110.1	112.6	117.1	117.6	120.1	122.6	123.1	125.1	127.6	130.1	132.6	133.6	135.1	136.6	142.1	142.6	145.1	150.1	157.6	165.1	180.1	195.1	210.1	1.05	9.50	10.00
109.3	111.8	116.3	116.8	119.3	121.8	122.3	124.3	126.8	129.3	131.8	132.8	134.3	135.8	141.3	141.8	144.3	149.3	156.8	164.3	179.3	194.3	209.3	1.05	10.00	10.50
108.6	111.1	115.6	116.1	118.6	121.1	121.6	123.6	126.1	128.6	131.1	132.1	133.6	135.1	140.6	141.1	143.6	148.6	156.1	163.6	178.6	193.6	208.6	1.05	10.50	11.00
	115.8			123.3								138.3					153.3		168.3				1.06	* 7.50	* 8.00
	115.0			122.5								137.5					152.5		167.5				1.06	* 8.00	* 8.50
	114.2			121.7								136.7					151.7		166.7				1.06	* 8.50	9.00
	118.8			126.3								141.3					156.3		171.3				1.07	* 5.60	* 6.00
	116.6			124.1								139.1					154.1		169.1				1.07	* 7.00	* 7.50
104.2	106.7	111.2	111.7	114.2	116.7	117.2	119.2	121.7	124.2	126.7	127.7	129.2	130.7	136.2	136.7	139.2	144.2	151.7	159.2	174.2	189.2	204.2	1.07	13.00	14.00
105.8	108.3	112.8	113.3	115.8	118.3	118.8	120.8	123.3	125.8	128.3	129.3	130.8	132.3	137.8	138.3	140.8	145.8	153.3	160.8	175.8	190.8	205.8	1.08	12.00	13.00
107.4	109.9	114.4	114.9	117.4	119.9	120.4	122.4	124.9	127.4	129.9	130.9	132.4	133.9	139.4	139.9	142.4	147.4	154.9	162.4	177.4	192.4	207.4	1.09	11.00	12.00
109.7	112.2	116.7	117.2	119.7	122.2	122.7	124.7	127.2	129.7	132.2	133.2	134.7	136.2	141.7	142.2	144.7	149.7	157.2	164.7	179.7	194.7	209.7	1.10	9.50	10.50
109.0	111.5	116.0	116.5	119.0	121.5	122.0	124.0	126.5	129.0	131.5	132.5	134.0	135.5	141.0	141.5	144.0	149.0	156.5	164.0	179.0	194.0	209.0	1.10	10.00	11.00
	119.6			127.1								142.1					157.1		172.1				1.11	* 5.00	* 5.60
	113.8			121.3								136.3					151.3		166.3				1.11	* 8.50	9.50
110.5	113.0	117.5	118.0	120.5	123.0	123.5	125.5	128.0	130.5	133.0	134.0	135.5	137.0	142.5	143.0	145.5	150.5	158.0	165.5	180.5	195.5	210.5	1.11	9.00	10.00
	114.6			122.1								137.1					152.1		167.1				1.12	* 8.00	9.00
98.7	101.2	105.7	106.2	108.7	111.2	111.7	113.7	116.2	118.7	121.2	122.2	123.7	125.2	130.7	131.2	133.7	138.7	146.2	153.7	168.7	183.7	198.7	1.12	16.00	18.00
	115.4			122.9								137.9					152.9		167.9				1.13	* 7.50	* 8.50
	116.2			123.7								138.7					153.7		168.7				1.14	* 7.00	* 8.00
107.8	110.3	114.8	115.3	117.8	120.3	120.8	122.8	125.3	127.8	130.3	131.3	132.8	134.3	139.8	140.3	142.8	147.8	155.3	162.8	177.8	192.8	207.8	1.14	10.50	12.00
101.9	104.4	108.9	109.4	111.9	114.4	114.9	116.9	119.4	121.9	124.4	125.4	126.9	128.4	133.9	134.4	136.9	141.9	149.4	156.9	171.9	186.9	201.9	1.14	14.00	16.00
109.3	111.8	116.3	116.8	119.3	121.8	122.3	124.3	126.8	129.3	131.8	132.8	134.3	135.8	141.3	141.8	144.3	149.3	156.8	164.3	179.3	194.3	209.3	1.15	9.50	11.00
	117.7			125.2								140.2					155.2		170.2				1.16	* 6.00	* 7.00
110.1	112.6	117.1	117.6	120.1	122.6	123.1	125.1	127.6	130.1	132.6	133.6	135.1	136.6	142.1	142.6	145.1	150.1	157.6	165.1	180.1	195.1	210.1	1.16	9.00	10.50
105.0	107.5	112.0	112.5	115.0	117.5	118.0	120.0	122.5	125.0	127.5	128.5	130.0	131.5	137.0	137.5	140.0	145.0	152.5	160.0	175.0	190.0	205.0	1.16	12.00	14.00
	113.4			120.9								135.9					150.9		165.9				1.17	* 8.50	10.00
	114.2			121.7								136.7					151.7		166.7				1.18	* 8.00	9.50
106.6	109.1	113.6	114.1	116.6	119.1	119.6	121.6	124.1	126.6	129.1	130.1	131.6	133.1	138.6	139.1	141.6	146.6	154.1	161.6	176.6	191.6	206.6	1.18	11.00	13.00
	119.3			126.8								141.8					156.8		171.8				1.19	* 5.00	* 6.00
	115.0			122.5								137.5					152.5		167.5				1.19	* 7.50	9.00
108.2	110.7	115.2	115.7	118.2	120.7	121.2	123.2	125.7	128.2	130.7	131.7	133.2	134.7	140.2	140.7	143.2	148.2	155.7	163.2	178.2	193.2	208.2	1.19	10.00	12.00
	115.8			123.3								138.3					153.3		168.3				1.20	* 7.00	* 8.50
109.7	112.2	116.7	117.2	119.7	122.2	122.7	124.7	127.2	129.7	132.2	133.2	134.7	136.2	141.7	142.2	144.7	149.7	157.2	164.7	179.7	194.7	209.7	1.21	9.00	11.00
	113.0			120.5								135.5					150.5		165.5				1.22	* 8.50	10.50
102.7	105.2	109.7	110.2	112.7	115.2	115.7	117.7	120.2	122.7	125.2	126.2	127.7	129.2	134.7	135.2	137.7	142.7	150.2	157.7	172.7	187.7	202.7	1.22	13.00	16.00
	118.1			125.6								140.6					155.6		170.6				1.23	* 5.60	* 7.00
	117.3			124.8								139.8					154.8		169.8				1.23	* 6.00	* 7.50
107.0	109.5	114.0	114.5	117.0	119.5	120.0	122.0	124.5	127.0	129.5	130.5	132.0	133.5	139.0	139.5	142.0	147.0	154.5	162.0	177.0	192.0	207.0	1.23	10.50	13.00
	113.8			121.3								136.3					151.3		166.3				1.24	* 8.00	10.00
97.2	99.7	104.2	104.7	107.2	109.7	110.2	112.2	114.7	117.2	119.7	120.7	122.2	123.7	129.2	129.7	132.2	137.2	144.7	152.2	167.2	182.2	197.2	1.24	16.00	20.00
	114.6			122.1								137.1					152.1		167.1				1.25	* 7.50	9.50
108.6	111.1	115.6	116.1	118.6	121.1	121.6	123.6	126.1	128.6	131.1	132.1	133.6	135.1	140.6	141.1	143.6	148.6	156.1	163.6	178.6	193.6	208.6	1.25	9.50	12.00
105.8	108.3	112.8	113.3	115.8	118.3	118.8	120.8	123.3	125.8	128.3	129.3	130.8	132.3	137.8	138.3	140.8	145.8	153.3	160.8	175.8	190.8	205.8	1.26	11.00	14.00
	115.4			122.9								137.9					152.9		167.9				1.27	* 7.00	9.00
	112.6			120.1								135.1					150.1		165.1				1.28	* 8.50	11.00
100.3	102.8	107.3	107.8	110.3	112.8	113.3	115.3	117.8	120.3	122.8	123.8	125.3	126.8	132.3	132.8	135.3	140.3	147.8	155.3	170.3	185.3	200.3	1.28	14.00	18.00
107.4	109.9	114.4	114.9	117.4	119.9	120.4	122.4	124.9	127.4	129.9	130.9	132.4	133.9	139.4	139.9	142.4	147.4	154.9	162.4	177.4	192.4	207.4	1.29	10.00	13.00
	113.4			120.9								135.9					150.9		165.9				1.30	* 8.00	10.50

Key to Horsepower Correction Factor: 0.7 | 0.8 | 0.9 | 1.0 | 1.1 | 1.2 | 1.3 | 1.4

We find that belt CP330 (330" perimeter) will provide a center distance of CD=146.6" for 12" diameter sheaves or CD=145"for 13" diameter sheaves. For this application 13"

sheaves may be a more conservative choice given our speed of 400 rpm. Hence 13" diameter sheaves are selected.

Belt speed = V = πdn = π(13/12)400 = 1361 fpm (feet/min) > 1000 fpm so OK.

Next we need to determine the Horsepower Correction Factor for our belt. This factor is the combination of Wrap and Belt Length Correction factors used in Mott and Shigley. From Table B23, we are located in the HP Correction value of 1.2 (light blue). This makes sense as the wrap angle value will be 1.0 for speed ratio of 1.00 (180° wrap angle) but the belt length will be higher than 1.0 given such a long center distance (12 feet). Shigley 17-14 has length factor of 1.2 for C belts of lengths 330" and up so this corresponds to the value on this table.

HP Correction value = 1.2

Next we need to find the rated horsepower per belt (H_{Tab}) for our application. We continue to scroll through the manual until we reach page B222 which has the power ratings for these hi power belts.

Table No. B33

C CP

Rated Horsepower per belt for C Section Predator V-Belts and Predator PowerBand Belts

RPM of Faster Shaft	Basic Horsepower per Belt for Small Sheave Datum Diameter									RPM of Faster Shaft	Additional Horsepower per Belt for Speed Ratio								
	9.00	9.50	10.00	10.50	11.00	12.00	13.00	14.00	16.00		1.00 to 1.01	1.02 to 1.02	1.03 to 1.04	1.05 to 1.05	1.06 to 1.07	1.08 to 1.10	1.10 to 1.13	1.14 to 1.19	1.20 to 1.28	1.29 and over
575	17.8	19.6	21.4	23.2	25.0	28.6	32.1	35.6	42.5	575	0.00	0.12	0.24	0.35	0.47	0.59	0.71	0.82	0.94	1.06
690	20.8	22.9	25.0	27.2	29.3	33.4	37.6	41.6	49.6	690	0.00	0.14	0.28	0.42	0.56	0.71	0.85	0.99	1.13	1.27
725	21.7	23.9	26.1	28.3	30.5	34.9	39.2	43.4	51.7	725	0.00	0.15	0.30	0.45	0.59	0.74	0.89	1.04	1.19	1.34
870	25.2	27.8	30.4	33.0	35.6	40.6	45.6	50.5	60.1	870	0.00	0.18	0.36	0.53	0.71	0.89	1.07	1.25	1.42	1.60
950	27.0	29.9	32.7	35.5	38.2	43.7	49.0	54.2	64.4	950	0.00	0.19	0.39	0.58	0.78	0.97	1.17	1.36	1.56	1.75
1160	31.7	35.0	38.3	41.5	44.8	51.1	57.3	63.3	74.8	1160	0.00	0.24	0.47	0.71	0.95	1.19	1.42	1.66	1.90	2.14
1425	36.9	40.8	44.6	48.4	52.1	59.4	66.4	73.1	85.7	1425	0.00	0.29	0.58	0.88	1.17	1.46	1.75	2.04	2.33	2.62
1750	42.4	46.9	51.2	55.5	59.6	67.6	75.2			1750	0.00	0.36	0.72	1.07	1.43	1.79	2.15	2.51	2.87	3.22
100	3.84	4.19	4.55	4.91	5.26	5.97	6.67	7.37	8.77	100	0.00	0.02	0.04	0.06	0.08	0.10	0.12	0.14	0.16	0.18
200	7.10	7.79	8.47	9.15	9.83	11.2	12.5	13.9	16.5	200	0.00	0.04	0.08	0.12	0.16	0.20	0.25	0.29	0.33	0.37
300	10.1	11.1	12.1	13.1	14.1	16.1	18.1	20.0	23.9	300	0.00	0.06	0.12	0.18	0.25	0.31	0.37	0.43	0.49	0.55
400	13.0	14.3	15.6	16.9	18.2	20.8	23.3	25.9	30.9	400	0.00	0.08	0.16	0.25	0.33	0.41	0.49	0.57	0.65	0.74
500	15.8	17.4	19.0	20.6	22.2	25.3	28.4	31.5	37.6	500	0.00	0.10	0.20	0.31	0.41	0.51	0.61	0.72	0.82	0.92
600	18.5	20.4	22.2	24.1	26.0	29.7	33.3	36.9	44.1	600	0.00	0.12	0.25	0.37	0.49	0.61	0.74	0.86	0.98	1.11
700	21.0	23.2	25.4	27.5	29.6	33.8	38.0	42.1	50.2	700	0.00	0.14	0.29	0.43	0.57	0.72	0.86	1.00	1.15	1.29
800	23.5	25.9	28.4	30.8	33.2	37.9	42.6	47.2	56.1	800	0.00	0.16	0.33	0.49	0.66	0.82	0.98	1.15	1.31	1.47
900	25.9	28.6	31.3	33.9	36.6	41.8	46.9	51.9	61.8	900	0.00	0.18	0.37	0.55	0.74	0.92	1.11	1.29	1.47	1.66
1000	28.2	31.1	34.1	37.0	39.8	45.5	51.1	56.5	67.1	1000	0.00	0.20	0.41	0.61	0.82	1.02	1.23	1.43	1.64	1.84
1100	30.4	33.6	36.7	39.9	43.0	49.0	55.0	60.8	72.0	1100	0.00	0.23	0.45	0.68	0.90	1.13	1.35	1.58	1.80	2.03
1200	32.5	35.9	39.3	42.6	45.9	52.4	58.8	64.9	76.6	1200	0.00	0.25	0.49	0.74	0.98	1.23	1.47	1.72	1.96	2.21
1300	34.5	38.2	41.7	45.3	48.8	55.6	62.3	68.7	80.9	1300	0.00	0.27	0.53	0.80	1.06	1.33	1.60	1.86	2.13	2.39
1400	36.4	40.3	44.1	47.8	51.5	58.7	65.6	72.3	84.7	1400	0.00	0.29	0.57	0.86	1.15	1.43	1.72	2.01	2.29	2.58
1500	38.3	42.3	46.3	50.2	54.0	61.5	68.7	75.5	88.2	1500	0.00	0.31	0.61	0.92	1.23	1.54	1.84	2.15	2.46	2.76
1600	40.0	44.2	48.4	52.4	56.4	64.1	71.5	78.4		1600	0.00	0.33	0.65	0.98	1.31	1.64	1.96	2.29	2.62	2.95
1700	41.6	46.0	50.3	54.5	58.6	66.5	74.0	81.1		1700	0.00	0.35	0.70	1.04	1.39	1.74	2.09	2.44	2.78	3.13
1800	43.2	47.7	52.1	56.4	60.6	68.7	76.3			1800	0.00	0.37	0.74	1.11	1.47	1.84	2.21	2.58	2.95	3.32
1900	44.6	49.2	53.8	58.2	62.5	70.7				1900	0.00	0.39	0.78	1.17	1.56	1.94	2.33	2.72	3.11	3.50
2000	45.9	50.6	55.3	59.8	64.1	72.4				2000	0.00	0.41	0.82	1.23	1.64	2.05	2.46	2.87	3.27	3.68
2100	47.0	51.9	56.6	61.2	65.6					2100	0.00	0.43	0.86	1.29	1.72	2.15	2.58	3.01	3.44	3.87
2200	48.1	53.1	57.8	62.4						2200	0.00	0.45	0.90	1.35	1.80	2.25	2.70	3.15	3.60	4.05
2300	49.0	54.1	58.9							2300	0.00	0.47	0.94	1.41	1.88	2.35	2.82	3.30	3.77	4.24
2400	49.8	54.9								2400	0.00	0.49	0.98	1.47	1.97	2.46	2.95	3.44	3.93	4.42
2500	50.5	55.6								2500	0.00	0.51	1.02	1.54	2.05	2.56	3.07	3.58	4.09	4.61
2600	51.0									2600	0.00	0.53	1.06	1.60	2.13	2.66	3.19	3.73	4.26	4.79
2700										2700	0.00	0.55	1.11	1.66	2.21	2.76	3.32	3.87	4.42	4.97
2800										2800	0.00	0.57	1.15	1.72	2.29	2.87	3.44	4.01	4.58	5.16
2900										2900	0.00	0.59	1.19	1.78	2.37	2.97	3.56	4.15	4.75	5.34
3000										3000	0.00	0.61	1.23	1.84	2.46	3.07	3.68	4.30	4.91	5.53
3100										3100	0.00	0.63	1.27	1.90	2.54	3.17	3.81	4.44	5.08	5.71
3200										3200	0.00	0.65	1.31	1.97	2.62	3.28	3.93	4.58	5.24	5.89
3300										3300	0.00	0.68	1.35	2.03	2.70	3.38	4.05	4.73	5.40	6.08

Drives for rpm-diameter combinations where no horsepower is shown may be practical if all conditions are known. See your local Gates representative

We find on page B230 the CP belts (Table B33). From the Table we see that for 13" diameter sheaves turning at 400rpm, the rated horsepower H_{Tab} = 23.3Hp.

We now apply the Horsepower correction factor to this value to get the Allowable Power per belt. The additional horsepower, ΔP is 0 given the speed ratio is 1.00

H_{all} = (HP Correction value) * (H_{Tab} + ΔP) =1.2(23.3 +0) = 27.96 HP

Number of belts = H_{design} / H_{Tab} = 90/27.96 = 3.22

Round up to 4 belts

A total of four CP330 belts are required for the pair of 13" sheaves at CD=145".

b) What is the factor of safety?

$n_{fs} = N(H_{all})/H_{des} = 4\ (27.96)/90 = 1.24$

Safety Factor = 1.24. This safety factor is adequate. We achieve a higher safety factor for 4 belts compared 5 belts in Problem 1.

Redoing the calculations from Problem 1. Eq. (17-22) the transmitted force and torque per belt are:

$$\Delta F_a = \frac{63\,025 H_a}{n(d/2)} = \frac{63\,025(18.63)}{400(26/2)} = 225.8 \text{ lbf/belt}$$

$$T_a = \frac{(\Delta F_a)d}{2} = \frac{225.8(26)}{2} = 2935 \text{ lbf} \cdot \text{in/belt}$$

The centripetal tension Eq. (17-21) is updated with $K_c = 1.716$ for Type C belts:

$$F_c = 3.498\left(\frac{V}{1000}\right)^2 = 3.498\left(\frac{2722.7}{1000}\right)^2 = 25.9 \text{ lbf/belt}$$

At fully developed friction, Eq. (17-9) gives

$$F_i = \frac{T}{d}\left[\frac{\exp(f\theta)+1}{\exp(f\theta)-1}\right] = \frac{2935}{26}\left(\frac{5+1}{5-1}\right) = 169.3 \text{ lbf/belt}$$

Eq. (17-10): $$F_1 = F_c + F_i\left[\frac{2\exp(f\theta)}{\exp(f\theta)+1}\right] = 25.9 + 169.3\left[\frac{2(5)}{5+1}\right] = 308.1 \text{ lbf/belt}$$

$$F_2 = F_1 - \Delta F_a = 308.1 - 225.8 = 82.3 \text{ lbf/belt}$$

$$n_f = \frac{H_a N_b}{H_d} = \frac{18.63(5)}{84} = 1.109 \quad \text{Ans.}$$

We have increased the transmitted forces in our belts to account for the lower sheave diameters.

c) What is the recommended static pre-tension for the belts? What should be the deflection at the mid-span to predict the correct pre-tension? Determine the recommended range of force at the mid-span to predict the correct pre-tension. Comment on the results.

Let's compare the pre-tension value to the Gates catalogue static pretension value. The calculations are found on page D24.

Regular V-Belt Tensioning Method

Step 1 **Calculate the Required Base Static Installation Tension Per Strand of Belt (Static Tension)**

A. The **static tension per strand** (T_{st}) is given by this formula:

Formula No. D19

$$T_{st} = 15\left(\frac{2.5^* - K\theta}{K\theta}\right)\left(\frac{(\text{Motor HP})\ (10^3)}{(N)(V)}\right) + \frac{MV^2}{10^5}$$

Where: Kθ = arc correction factor from Table No. D26 or Table No. D11 on Page D12 for V-Flat drives.
N = Number of belts. (This is the number of strands in the case of PowerBand® Belts.)
V = Belt speed, ft./min.
M = Constant from Table No. D27.
*2.67 for Micro-V® Belts.

Table No. D26

Arc of Contact Correction Factor Kθ for V-V Drives

$\frac{D-d}{C}$	Arc of Contact on Small Sheave (°)	Factor Kθ	
		A, B, C, D 3V, 5V, 8V 5M, 7M, 11M	Micro-V J, L, M
0.00	180	1.00	1.00
0.10	174	0.99	0.98
0.20	169	0.97	0.97
0.30	163	0.96	0.95
0.40	157	0.94	0.94
0.50	151	0.93	0.92
0.60	145	0.91	0.90
0.70	139	0.89	0.88
0.80	133	0.87	0.85
0.90	127	0.85	0.83
1.00	120	0.82	0.80
1.10	113	0.80	0.77

Table No. D27

Factor M and Factor Y

Cross Section	M	Y
Super HC® Molded Notch		
3VX	0.29	4
5VX	0.78	13
Super HC® Molded Notch PowerBand®		
3VX	0.39	4
5VX	0.98	13
Super HC		
5V	1.0	11
8V	2.6	22
Super HC PowerBand		
3V	0.46	4
5V	1.2	11
5VP	1.2	39
8V	3.0	22
Hi-Power® II		
A	0.51	7
B	0.80	8
C	1.5	18
D	3.0	27
Hi-Power II PowerBand		
A	0.66	7
B	1.0	9
C	1.8	18
D	3.4	28

Cross Section	M	Y
Tri-Power® Molded Notch		
AX	0.47	7
BX	0.76	8
CX	1.31	15
Micro-V® Belt		
J*	0.035	0.56
L	0.130	1.90
M	0.520	6.30
Polyflex® JB®		
5M**	0.05	1.2
7M	0.14	4.6
11M	0.31	8.5
Predator Singles		
AP	0.56	21
BP	0.84	38
CP	1.6	87
3VP	NA	NA
5VP	0.96	42
8VP	2.5	105
Predator PowerBand		
AP	NA	NA
BP	1.0	40
CP	1.6	89
3VP	0.46	8.8
5VP	1.2	39
8VP	3.0	105

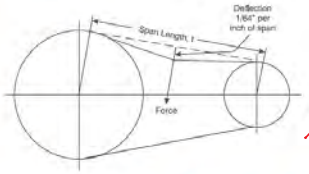
NOTE: When applying static belt tension values directly, multiply the required base static installation tension(Tst) calculated in Formula D19 by the following factors:

For New Belts:
Minimum Static Tension = 1.0 x Tst
Minimum Static Tension = 1.1 x Tst

For Used Belts:
Minimum Static Tension = 0.7 x Tst
Minimum Static Tension = 0.8 x Tst

Step 2 **Calculate the Minimum and Maximum Recommended Forces to Deflect One Belt 1/64" Per Inch of Span Length**

A. Measure the **span length** (t) of your drive (see sketch).



not correct

$t = \sqrt{C^2 - \left(\frac{D-d}{2}\right)^2}$

Figure No. D25

B. If your drive uses **two or more** PowerBand Belts or individual belts, calculate the lower and upper recommended deflection forces by these formulas:

Formula No. D20

$$\text{Minimum Recommended Force} = \frac{1.4\ T_{st} + Y}{16}$$

Formula No. D21

$$\text{Maximum Recommended Force} = \frac{1.5\ T_{st} + Y}{16}$$

Where: T_{st} = tension per strand from Step 1.
Y = constant from Table No. D27.

C. If your drive has only **one** PowerBand Belt (See Step D) or individual belt, calculate the lower and upper recommended deflection forces by these formulas:

Formula No. D22

$$\text{Minimum Recommended Force} = \frac{1.4\ T_{st} + \left(\frac{t}{L}\right)Y}{16}$$

Formula No. D23

$$\text{Upper Recommended Force} = \frac{1.5\ T_{st} + \left(\frac{t}{L}\right)Y}{16}$$

Where: T_{st} = tension per strand from Step 1.
Y = constant from Table No. D27.
t = span length (see Figure No. D25).
L = belt length

D. The deflection forces calculated in Step 2B or 2C are for an individual belt. Multiply these forces by the number of individual strands in a band to get the lower and upper recommended forces for a PowerBand Belt. (If your drive uses 2 or more PowerBand Belts, use the band with the fewest number of strands.)

From Table D27 for CP belts we have constants M=1.6 and Y = 87. Recall for 180° wrap angle K_ϕ = 1.0. The motor HP = 60, N= 4 belts, V = 1361fpm

T_{st} = 15((2.5-1/1))(60*1000/(4*1361)) +(1.6*(1361)^2)/1000000 = 251 lbf.

This value is lower than what was calculated using the Shigley equations (F_i = 409 lbf). The main difference is related to the fact that Shigley equation uses the design horsepower (including service factor) while the nominal power is input into Equation D19 (it is stationary so there can no shock loads). In addition, equation D19 uses curve fit through experimental data (always has some error) and perhaps an incorrect value for friction coefficients for the Shigley equations.

If the design power is used in Equation D19 we get:

T_{st} = 15((2.5-1/1))(90*1000/(4*1361)) +(1.6*(1361)^2)/1000000 = 375 lbf.

However, the true value is 251 lbf for the deflection calculations.

Lastly, the calculations for deflection of the belt to reflect the proper pre-tension.

The Span Length t = Center Distance for speed ratio = 1.00 (Wrap angle = 180°)

Span Length t = CD = 145”
Deflection for required force is t *1/64” = 145/64” = 2.266” = approx.2.25”

Forces required are from equations D20 and D21 (two or more belts). Recall Y = 87 from Table D27

Min. force required = ($1.4T_{st}$ + Y)/16 = (1.4(251) +87)/16 = 27.4 lbf
Max. force required = ($1.5\ T_{st}$ + Y)/16 = (1.5(251)+87)/16 = 29.0 lbf

Hence deflecting the center of the span 2.25” with a force of about 28lbf will ensure proper belt tension and belt life.

3. Adapted from Shigley, Problem 17-29.

A mine hoist uses a 2-inch 6 × 19 monitor-steel wire rope. The rope is used to haul maximum loads of 4 tons from a 480 ft. shaft. The drum has a diameter of 6 feet and the sheaves are of good-quality cast steel, and the smallest is 3 feet in diameter.

Part 1: To be demonstrated by the TA.

a. Using a maximum hoisting speed of 1200 feet/minute and a maximum acceleration of 2 ft/s^2, estimate the stresses in the wire rope.

The objective of the problem is to explore factors of safety in wire rope. We will express strengths as tensions. There are three conditions to consider:

1. Wire Rope Tension limits due to loads and acceleration/decelerations F_t
2. Wire Rope Tension limits due to bending through the sheave F_b
3. Wire Rope Tension limits due to fatigue F_f

Given: Monitor steel 2-in diameter 6 × 19 wire rope, 480 ft long.

From Table 17-27:
Recommended Minimum diameter of a sheave is 30*d*.
To allow a greater factor of safety we can use the "better sheave diameter," which is 45*d*.
Hence, 45(2") = 90 inch sheave diameter is selected. The hoist abuses the wire when it is bent around a sheave.

From Table 17-24:
The nominal tensile strength for 6 × 19 wire rope made from Monitor Steel is 106 kpsi (106,000 lbf/in^2). This is from Table 17-24. The value is lower than the values found on page 922 as it is applied to the entire area ($\pi d^2/4$) and not the actual area of metal. We define it as the nominal strength based on total area.

Hence, the ultimate load is:
Note 1 kip (kilopound) = 1000 lbf

$$F_u = (S_u)_{\text{nom}} A_{\text{nom}} = 106\left[\frac{\pi(2)^2}{4}\right] = 333 \text{ kip} \quad Ans.\ (333{,}000 \text{ lbf})$$

This breaking strength is very high 167.5 tons!

Wesco Industries Ltd.
PowerStrand Wire Rope



6 x 19 FC Class Wire Rope

6 Outer Strands, Rotation Wire Rope

Includes 6x19S & 6x26WS construction types. A good general purpose wire rope which is wear resistant and suitable for sling ropes, marine rope, lashing, fishing, anchor lines and logging.
6x19S (9+9+1) FC. Load bearing outer wires 114
6x26WS (10+(5+5)+5+1) FC. Load bearing outer wires 156
Fill factor 0.49
Use of swivel is not permitted

Diameter Inches	Weight per Foot Pounds	Grade Tensile	Bright		Galvanized	
			Wesco Code	Minimum Strength Pounds	Wesco Code	Minimum Strength Pounds
1/4	0.11	EIPS	W61908C1	8,040	W61908D1	5,436
5/16	0.16	EIPS	W61910C1	9,380	W61910D1	8,442
3/8	0.24	EIPS	W61912C1	13,420	W61912D1	12,078
7/16	0.32	EIPS	W61914C1	18,180	W61914D1	16,362
1/2	0.42	EIPS	W61916C1	23,600	W61916D1	21,240
9/16	0.53	EIPS	W62616C1	29,800	W61918D1	26,820
5/8	0.66	EIPS	W62620C1	36,600	W61920D1	32,940
3/4	0.95	EIPS	W62624C1	52,400	W61924D1	47,160
7/8	1.29	EIPS	W62628C1	70,800	W61928D1	63,720
1	1.68	EIPS	W62632C1	92,000	W61932D1	82,800
1-1/8	2.13	EIPS	W62636C1	115,800	W61936D1	104,220
1-1/4	2.63	EIPS	W62640C1	142,000	W61940D1	127,800
1-1/4	2.63	EEIPS	--	--	W62640D1C-EEIPS	156,200
1-3/8	3.18	EIPS	W62644C1	170,800	W61944D1	153,720
1-1/2	3.78	EIPS	W62648C1	202,000	W61948D1	181,800
1-5/8	4.4	EIPS	W62652C1	236,000	W61952D1	212,400
1-3/4	5.15	EIPS	W62656G1	272,000	W61956D1	244,800
1-7/8	5.91	EIPS	W62660G1	310,000	W61960D1	279,000
2	6.72	EIPS	W62664C1	352,000	W61964D1	316,800
2-1/4	8.51	EIPS	W62672C1	440,000	W61972D1	396,000
2-1/2	10.50	EIPS	W62680C1	538,000	W62680D1	484,200

EIPS – Extra Improved Plow Steel

EEIPS – Extra, Extra, Improved Plow Steel



Comparison of our computed value with published values is reasonable. Newer wire ropes are using Extra Improved Plow Steel are giving breaking strengths of 352,000 lbs for 2" rope (176 tons). The steel is slightly better than the one used in our calculations.

Next, we will compare this value with our working loads to determine the safety factors.

1. The tensile loading of the wire is given by Eq. (17-47)

The wire rope supports a load of 4 tons. Recall 1 ton = 2000 lbf

$$F_t = \left(\frac{W}{m} + wl\right)\left(1 + \frac{a}{g}\right)$$

$$W = 4(2) = 8 \text{ kip}, \quad m = 1$$

W = weight at end of rope (lbf)
m = number of wire ropes supporting the load
w = weight/foot of the wire rope (lbf/ft)
l = maximum length (ft)
a = maximum expected acceleration/deceleration (ft/sec^2)
g = gravity acceleration = 32.2 ft/sec^2

From Table (17-24), the weight/foot × length gives us:

$$wl = 1.60d^2 l = 1.60(2^2)(480) = 3072 \text{ lbf} = 3.072 \text{ kip}$$

Therefore,

$$F_t = (8 + 3.072)\left(1 + \frac{2}{32.2}\right) = 11.76 \text{ kip} \quad Ans.$$

2. Allowable bending load (tension) in wire rope. We use equation 17-41

From Eq. (17-41) and Table 17-27 we find the *equivalent bending load*, F_b as:

$$F_b = \frac{E_r d_w A_m}{D}$$

For 6×19 wire rope made from Monitor Steel the *modulus of elasticity for the rope* is E_r = 12 Mpsi = 12 (10^6) psi (Table 17-24 and Table 17-27)

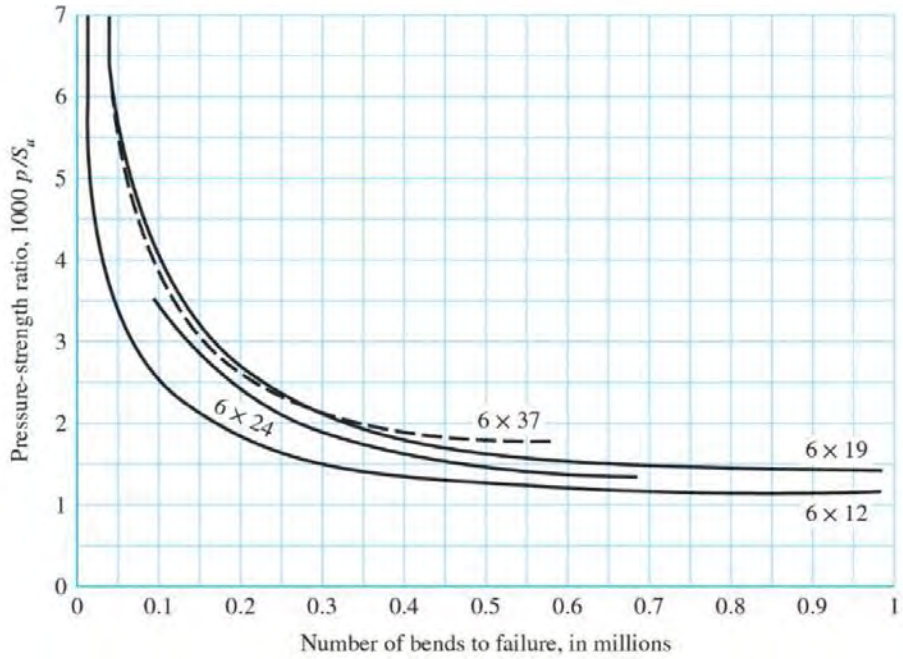
Table 17-27 Wire diameter $d_w = 0.067d$, Area of metal $A_m = 0.40d^2$

With a 90-in drum, we get:

$$F_b = 12(10^6)(0.067d)(0.40d^2)/D$$

$$F_b = 12(10^6)(0.067(2))(0.40(2^2)/90 = 28.6 \text{ kip} \quad Ans.$$

3. Allowable load based on fatigue requires the use of Figure 7-21 which provides an endurance limit which is given as a pressure-strength ratio. Ideally we want to predict the life of the wire rope to last more than 1,000,000 bend cycles (infinite life). Hence we pick the lowest ratio for the type of wire rope



From Figure 17-21 for 6 by 19 wire rope @10^6 bends we see that p/S_u = 1.4/1000= 0.014.

This value is used in eq'n 7-44 to predict the *allowable fatigue tension* F_f in the wire rope

$$F_f = \frac{(p/S_u)S_u dD}{2} \qquad \textbf{(17 - 44)}$$

From page 922 we have ranges for ultimate strength for the individual steel wires. In our case:

Improved Plow Steel (monitor) $240 < S_u < 280$ kpsi
We choose the lower value in this range.
With Sheave diameter = 90 inches we get:

$$(p / S_u) = 0.0014$$
$$S_u = 240 \text{ kpsi}, \quad \text{p. 920}$$
$$F_f = \frac{0.0014(240)(2)(72)}{2} = 24.2 \text{ kip} \quad Ans.$$

90
30.3

Tension Load = 11,760 lbs., Bending Load = 28600 lbs, and Fatique Load = 30300 lbs.
Ultimate Load = 333,000 lbs

To be demonstrated by the TA.

b. Estimate the various factors of safety in the wire rope.

1. *Static, no bending:*

$$n = \frac{F_u}{F_t} = \frac{333}{11.76} = 28.3 \quad Ans.$$

2. *Static, with bending:* 28.6

Eq. (17-46): $$n_s = \frac{F_u - F_b}{F_t} = \frac{333 - 39}{11.76} = \cancel{25.0} \quad Ans.$$ 25.9

3a. Fatigue without bending: 30.3

$$n_f = \frac{F_f}{F_t} = \frac{\cancel{34.2}}{11.76} = \cancel{2.06} \quad Ans.$$ 2.58

3b. Fatigue, with bending:

$$n_f = \frac{F_f - F_b}{F_t} \qquad \textbf{(17-45)}$$

$n_f = 30.3\text{-}28.6/11.76 = 0.14$

We expect bending to occur so 3b is the true safety factor for fatigue. Our value of 0.14 is < 1.0 and so this indicates that the wire rope selected is not suitable for this application for an infinite life expectancy. We would need to consider using different configurations if we want to try to get higher safety factors in fatique.

One approach is if we study the idea of replacing the wire after 100,000 cycles. In this case we have the following:

For a life of $0.1(10^6)$ cycles, from Fig. 17-21 we get:

$$(p / S_u) = 4 / 1000 = 0.004$$

$$F_f = \frac{0.004(240)(2)(\cancel{72})}{2} = \cancel{69.1}\text{kip}$$ 90, 86.4 kip

86.4 28.6

Eq. (17-45): $$n_f = \frac{\cancel{69.1} - 39}{11.76} = \cancel{2.56} \quad Ans.$$ 4.91

This can now incorporate a scheduled replacement for the wire ropes. Say 100 operations a day is expected. This will give 1000 days or about 3 years before the cable must be replaced.

Comments:

- There are a number of factors of safety used in wire rope analysis. They are different, with different meanings. There is no substitute for knowing exactly which factor of safety is written or spoken.
- Static performance of a rope in tension is impressive.
- The remedy for fatigue is the use of smaller diameter ropes, **with multiple ropes supporting the load**. See Ex. 17-6 for the effectiveness of this approach. It will also be used in Prob. 17-30.
- Remind students that wire ropes do not fail suddenly due to fatigue. The outer wires gradually show wear and breaks; such ropes should be retired. Periodic inspections prevent fatigue failures by parting of the rope.